FLAT MORPHISMS OF RELATIVE DIMENSION R RAISE THE DELTA-DIMENSION OF CLOSED SUBSETS BY R

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a reduction to an integral base and one component of the preimage, openness and dominance of that component, and a fibre dimension computation at the generic points

Editorial note. The label lemma-flat-inverse-image-dimension names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The delta-dimension of a closed subset as the supremum of the delta-dimensions of its irreducible components, and the delta-dimension of a point
- Openness of a flat morphism that is locally of finite presentation, and the criterion placing generic points in the schematic locus
- The dimension formula for the fibre of a morphism at a point, and the definition of a morphism of a given relative dimension

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 1150; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04NQ: sources/stacks-chow/tags.txt, source lines 6088–6088. Prerequisite pointer invoked at proof line(s) 1186; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 042S: sources/stacks-chow/tags.txt, source lines 5339–5339. Prerequisite pointer invoked at proof line(s) 1172; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 06NH: sources/stacks-chow/tags.txt, source lines 8536–8536. Prerequisite pointer invoked at proof line(s) 1180; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *In Situation 02QL let $X, Y/B$ be good. Let $f : X \to Y$ be a morphism over B. Assume f is flat of relative dimension r. For any closed subset $T \subset |Y|$ we have*

$$\dim_\delta(|f|^{-1}(T)) = \dim_\delta(T) + r.$$

provided $|f|^{-1}(T)$ is nonempty. If $Z \subset Y$ is an integral closed subscheme and $Z' \subset f^{-1}(Z)$ is an irreducible component, then Z' dominates Z and $\dim_\delta(Z') = \dim_\delta(Z) + r$.

Proof. Since the δ-dimension of a closed subset is the supremum of the δ-dimensions of the irreducible components, it suffices to prove the final statement. We may replace Y by the integral closed subscheme Z and X by $f^{-1}(Z) = Z \times_Y X$. Hence we may assume $Z = Y$ is integral and f is a flat morphism of relative dimension r. Since Y is locally Noetherian the morphism f which is locally of finite type, is actually locally of finite presentation. Hence Morphisms of Spaces, Lemma 042S applies and we see that f is open. Let $\xi \in X$ be a generic point of an irreducible component of X. By the openness of f we see that $f(\xi)$ is the generic point η of $Z = Y$. Thus Z' dominates $Z = Y$. Finally, we see that ξ and η are in the schematic locus of X and Y by Properties of Spaces, Proposition 06NH. Since ξ is a generic point of X we see that $\mathcal{O}_{X,\xi} = \mathcal{O}_{X_\eta,\xi}$ has only one prime ideal and hence has dimension 0 (we may use usual local rings as ξ and η are in the schematic loci of X and Y). Thus by Morphisms of Spaces, Lemma 04NQ (and the definition of morphisms of given relative dimension) we conclude that the transcendence degree of $\kappa(\xi)$ over $\kappa(\eta)$ is r. In other words, $\delta(\xi) = \delta(\eta) + r$ as desired. □